Din Sopheak Panha

Robotics and Automation Engineer
Autonomous Mobile Robot Navigation and Path Tracking

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RESEARCH INTERESTS

Autonomous mobile robots, navigation, path tracking, teleoperation, shared control under communication delay, human-robot collaboration.

EDUCATION

2021 to 2026 **Engineer's Degree in Robotics and Automation Engineering**, Institute of Technology of Cambodia, Phnom Penh, Cambodia

Department of Industrial and Mechanical Engineering, ECAM Engineering double degree program with ECAM LaSalle (France), 2024 to 2026

GPA: 3.5 / 4.0 overall; by year: 3.5 (Year 1), 3.5 (Year 2), 4.0 (Year 3), 3.5 (Year 4), 3.5 (Year 5)

Thesis: Implementation of Nav2 for Autonomous Indoor Delivery Robot, defended July 2026

Language of instruction: English

Relevant coursework (all grade A): AI and Machine Learning, Introduction to AI, Data, Automation and Industrial Robotics, Sensing and Perception, Introduction to Control Theory, Feedback Control Systems, Signals and Systems, Microprocessor Architecture, Computer Programming, Calculus I, II and III, Differential Equations, Linear Algebra, Probability, Statistics, Numerical Methods and Optimization

COMPETITIONS, LEADERSHIP & MEMBERSHIPS

2026-Present **Member**, Union of Youth Federations of Cambodia (UYFC)

2025 **Team Leader, ITC01**, ABU Robocon 2025, GEE Lab (Electronics Engineering Lab), Institute of Technology of Cambodia. Led the mechanical, electronics and software sub-teams, programmed both robots together with a teammate, and assisted in designing the motor driver, controller shield and pneumatic valve PCBs for the CAN-based electronics.

2024 **Electronics Team Member, ITC02**, ABU Robocon 2024, Dynamics and Control Laboratory (DCLab), Institute of Technology of Cambodia. Assisted with PCB assembly, soldering, robot wiring and hardware integration.

2023-Present **Member**, Dynamics and Control Laboratory (DCLab)

CONFERENCE PUBLICATIONS

2026 **Sopheak Panha Din**, Sarot Srang, Sela Van. Design and integration of a lookahead-based path tracking system for a three-wheel swerve-drive robot using ROS 2. *6th International Conference on Robotics, Automation, and Artificial Intelligence (RAAI 2026)*, Singapore, 2026. Under review

RESEARCH EXPERIENCE

2026 to present · Active Projects

- **Shared control teleoperation under communication delay (independent research, in progress)**: when an operator drives a mobile robot remotely, network delay makes commands and sensor data arrive late and collisions more likely. Developing shared control, where the robot adds collision-avoidance assistance to the operator's commands, that still works under delay, in a ROS 2 and Gazebo simulation.
- **DIY dual-axis solar tracker with energy monitoring (STEAMING Cambodia, in progress)**: a panel that follows the sun on two axes by comparing the light on four sensors and turning toward the brighter side, while measuring the panel's voltage, current and power on a web dashboard.

2025 to 2026 · Institute of Technology of Cambodia, Dynamics and Control Laboratory (DCLab)

- **Lookahead-based path tracking for a three-wheel swerve-drive robot (2025)**: PID path tracking on a swerve-drive robot needs separate controllers for x, y and heading that must be retuned for each path. Designed a lookahead-based controller that drives the robot toward a point ahead on the path as a velocity vector, with feedforward heading control and a corner-aware speed limit, using LiDAR-inertial localization. It tracked square, circle and triangle paths with RMS cross-track errors of 0.77 to 1.09 cm (2.89 cm on a star path) at up to 1.5 m/s, with one fixed parameter set.

- **Autonomous indoor delivery robot (thesis, 2026)**: implemented the Nav2 navigation stack so a differential-drive robot maps a classroom, localizes itself and drives to a goal while avoiding obstacles that were not in the map, using SLAM mapping, particle filter localization (AMCL), a Smac 2D planner, a Vector Pursuit path tracking controller and keepout zones. It reached the goal in all 10 trials with no collisions, with a mean position error of 0.256 m without and 0.260 m with added obstacles; the remaining error was traced to localization drift rather than control.

2025 · ECAM LaSalle, France (exchange semester)

- **Face tracking teleoperation with SO-ARM101**: hands-free control of a robot arm without the usual second leader arm. Facial landmarks from a single webcam map head yaw, head pitch and mouth opening to the base, wrist and gripper at 60 Hz, with a deadband that removes head jitter so the arm holds still. Led the team as Scrum Master.

2025 · Machine Learning

- **Child malnutrition prediction from household survey data**: predicted which households have child malnutrition from 17 living-condition indicators (electricity, water, sanitation, education, region) in the Cambodia Demographic and Health Survey (9,099 households). As only 25% are malnourished, neural networks were trained with class weights and judged by recall and ROC-AUC, not accuracy. The best model detected 82% of malnourished households (ROC-AUC 0.78); K-Means clustering could not separate them.

2026 · Industrial Automation

- **Sequence variation system with an ABB robot arm**: a pick-and-place cell where the robot takes objects from a 4x4 grid and places them at two stations in turn, in a continuous loop. All 16 pick positions are computed from one taught target with offsets, a cycle counter selects the station, and waiting for the vacuum pressure signal before lifting fixed failed grips. Also built a conveyor sorting system with a Siemens PLC and HMI (TIA Portal V18).

PROFESSIONAL EXPERIENCE

2026 to 2027	**Robotics Tutor**, STEAMING Cambodia (Nov 2026 to Feb 2027). Built the 12-session Micro:Bit Innovators Program single-handedly and teach a Robotics and IoT class with the ESP32.
2026	**Relay Protection Testing Engineer**, SNKRP (for EDC) (Aug 2026 to Oct 2026). Supported senior engineers in testing substation protection relays with an OMICRON CMC 356 and helped prepare test reports.
2026	**Robotics Tutor**, Seksaa Tech Academy (Jun 2026 to Aug 2026). Taught electronics and micro:bit programming to young learners.
2025 to 2026	**Robotics Engineer (part-time)**, Happy Sprouts (Sep 2025 to Jul 2026). Designed the electrical system for a 6-DOF mushroom-harvesting robot prototype (gantry with a SCARA arm), integrating ROS 2 with STM32 controllers and closed-loop stepper drivers over CAN bus. The prototype moves as designed and automatic harvesting is not yet complete.
2025	**Engineering Intern**, Startec Dynamics (Calypso Motors) (Jun 2025 to Sep 2025). Built a GPS, IMU and ESP32 motorbike tracking unit that detects start, stop and possible accidents.
2023	**Engineering Intern**, AI Farm (Aug 2023 to Oct 2023). Designed closed-loop stepper motor control for a robot arm on STM32, using PID with encoder feedback.

SKILLS

Programming	C/C++, Python, MATLAB
Robotics & Simulation	ROS 2 (Humble), Nav2, MoveIt 2, SLAM Toolbox, FAST-LIO2, Gazebo, RViz2
Vision & AI	YOLO, OpenCV, dlib, PyTorch, scikit-learn
Industrial Automation	Siemens TIA Portal (V18), ABB RobotStudio, PLC Programming, HMI Design, Ladder Logic, RAPID
Hardware & Embedded	3D LiDAR (Livox Mid-360), 2D LiDAR, Depth Cameras (Intel RealSense), IMU, CAN bus, NVIDIA Jetson, STM32, ESP32, Raspberry Pi/Orange Pi, Arduino
Tools & OS	Linux (Ubuntu), LaTeX, AutoCAD
Languages	Khmer (native), English (IELTS 6.0, Competent User, July 2022), French (intermediate)

REFERENCES

Asst. Prof. Dr. Sarot Srang, Head of DCLab and ECAM Engineering Program, ITC · srangsarot@itc.edu.kh

Mr. Kann Teudom, Head of STEM Education, STEAMING Cambodia · pdomsc@gmail.com